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A List of Prompts

The following prompts in Figure 4 are used with GPT-4o to produce each of the seven meaning-preserving perturbations described in § 3.1. Each prompt instructs the model to rewrite the summary according to a specific linguistic transformation while preserving factual meaning.

B Dataset Statistics

The detailed statistics for the datasets used in our experiments can be seen in Table 2.

C Faithfulness of Perturbed Summaries

To verify that the applied perturbations preserve factual consistency, we perform an automatic faithfulness check using an NLI-based approach. This analysis serves as a sanity check to ensure that the perturbations do not introduce widespread factual errors, rather than as a definitive evaluation of summary correctness.

For each perturbed summary, we split its text into sentences, evaluate it against the corresponding original summary, treating the original summary as the premise and the perturbed sentence as the hypothesis. If a premise–hypothesis pair exceeds the model's maximum input length, we apply sentence-level chunking to the premise and aggregate predictions across chunks (Scirè et al., 2024; Yang et al., 2024). A sentence is counted as contradictory if the NLI model[4] predicts a contradiction label. The contradiction rate for a summary is defined as the fraction of its perturbed sentences labeled as contradictory, and dataset-level results are obtained by averaging these rates across summaries. The resulting contradiction rates for each dataset and perturbation type are reported in Table 3. Illustrative examples of original and perturbed summaries for selected perturbation types are shown in Figures 5, 6, & 7.

Across most perturbations, contradiction rates remain low, indicating that paraphrasing, simplification, vocabulary reduction, summarization, and source text insertion generally preserve factual consistency with respect to the original summaries. This supports our assumption that score changes observed in the main experiments primarily reflect metric sensitivity to surface and structural variation, rather than systematic factual errors introduced by the perturbations.

[4] https://huggingface.co/MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli

We observe higher contradiction rates for the *Negated* perturbation across all datasets. This behavior is expected and does not necessarily indicate that these perturbations introduce factual errors. The *Negated* perturbation is explicitly designed to negate some statements in the summary, and therefore, a non-zero rate of contradictions indicates that the perturbation is being applied as intended. Importantly, our perturbation setup operates on the full summary as input, allowing the model to perform transformations at the summary level rather than strictly applying simple double negation to individual sentences. As a result, negation may be distributed across clauses or introduced through structural rewrites that preserve the overall meaning of the summary, but appear contradictory when evaluated sentence by sentence. Since our NLI-based validation operates at the sentence level, it may flag such locally negated sentences as contradictions even when the global summary meaning remains logically equivalent. Furthermore, the NLI-based validation used here is not designed to distinguish between *intended* negations that preserve overall factual meaning and *undesired* negations that fundamentally alter the factual content of the summary. Determining whether a negation invalidates the summary would require verifying each negated statement against the original source document, which in turn would necessitate fine-grained human evaluation over long inputs. Such an analysis is substantially more expensive and complex and lies beyond the scope of this work. As a result, higher contradiction rates for negated summaries should be interpreted as evidence that the perturbation successfully introduces negation, rather than as definitive proof of factual inconsistency. This limitation further highlights the need for more nuanced evaluation methods, including human verification, when assessing logical transformations in long-document summarization.

D Per-Dataset Results

Tables 4, 5, and 6 report mean factuality scores for each metric under all seven perturbation types and for the original summaries across the three datasets used in this study. These tables provide the complete quantitative results corresponding to the aggregate trends shown in Figure 2. Consistent with our main analysis, `MiniCheck` and `UniEval` appear to be the most robust overall, maintaining relatively stable scores across most perturbations

P1. Paraphrased
system_prompt: Provide the paraphrased version of the text.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P2. Less Diverse
system_prompt: Rewrite the following text using less diverse vocabulary.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P3. Negated
system_prompt: Rewrite the following text by introducing logically equivalent negations while preserving its original meaning.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P4. Simplified
system_prompt: Rewrite the following text by making complex sentences simpler.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P5. Summarized
system_prompt: Rewrite the text to make it more concise.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P6. Synonym Replacement
system_prompt: Revise the text using synonyms for some common words.\n\nYou are strictly prohibited from omitting any information or altering its original meaning. Do not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary>
P7. Added Source Text
system_prompt: Insert a source sentence into the summary that does not relate to its main ideas.\n\nDo not include explanations, reasoning, or commentary in your output.
user_prompt: Text: <summary> \n\n Source: <document>

Figure 4: Prompt templates used with GPT-4o to generate meaning-preserving perturbations of the original summaries.

Dataset	#Examples (Used)	Avg. Summary Sentences	Avg. Summary Tokens	Avg. Document Sentences	Avg. Document Tokens	Summary Type
SQuALITY	260	12.5	273	456.6	6,131	Human-written
LexAbSumm	351	4.2	169	385.9	10,840	Human-written
ScholarQABench	100	43.2	1,158	575.4	14,652	Human-written

Table 2: Dataset statistics for the three long-document summarization benchmarks used in this study.

and datasets, with the exception of degraded performance on *Negated* summaries. In contrast, the remaining metrics are influenced by almost all types of perturbations, showing greater score variability, particularly in the legal domain.

`BARTScore`[5] performs poorly even on the original (unperturbed) summaries across all datasets. As a generation-based metric, its scoring depends heavily on the size and structure of the retrieved context, which may not be the ideal case in long-document setting, where evidence for a single summary sentence may be scattered across distant sections of the source. The mismatch between the localized retrieved snippets and the broader document context can distort likelihood estimates, and this effect compounds when scores are aggregated over full summaries, leading to consistently lower values. We also observe that while a few individual sentences receive high `BARTScore` values, most have extremely low scores due to being more compressed and contextually demanding, which drives the overall average down.

E Domain-Specific Robustness

To further analyze how text characteristics across domains influence factuality metric robustness, we compute the mean absolute score change under meaning-preserving perturbations. While signed

[5]We use the implementation provided by Bishop et al. (2024).